

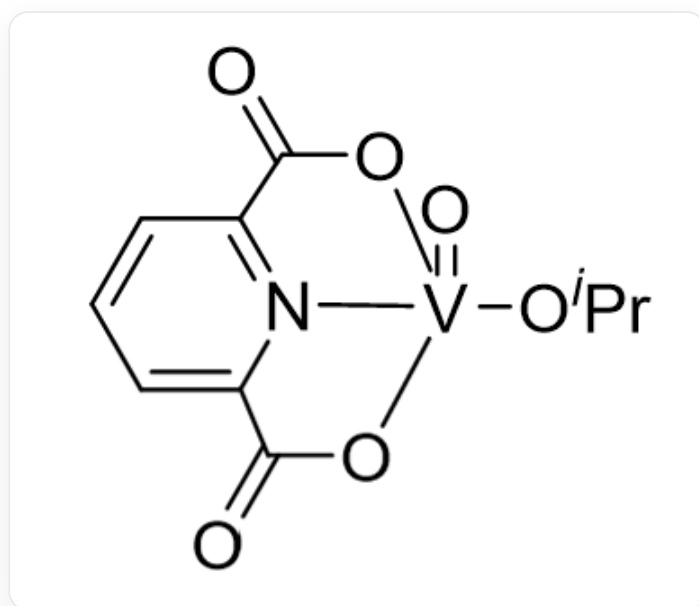
Question

Perdeuterated pyridine ($py - d_5$, denoted directly as py or Py in responses) can catalyze the pyrolysis of the vanadium compound $[\mathbf{L}(O^iPr)V=O](\mathbf{A})$ (where $\mathbf{L}^{2-}$ represents the 2,6-pyridinedicarboxylate anion). It is known that this reaction proceeds quantitatively and yields the following products (unbalanced):

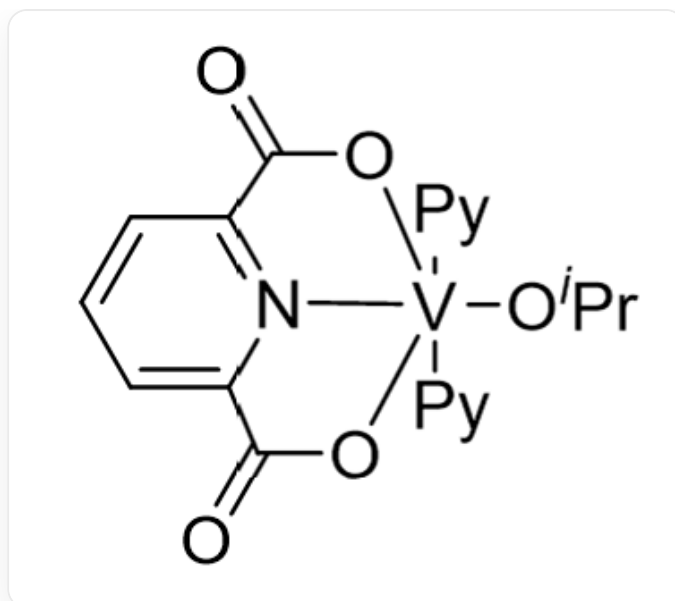


The structures of $\mathbf{A}$ and $\mathbf{P}$ are as follows:

$\mathbf{A}$:



$\mathbf{P}$:



The experimental team proposed three possible reaction pathways for the pyrolysis:

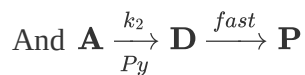
PathI



PathII

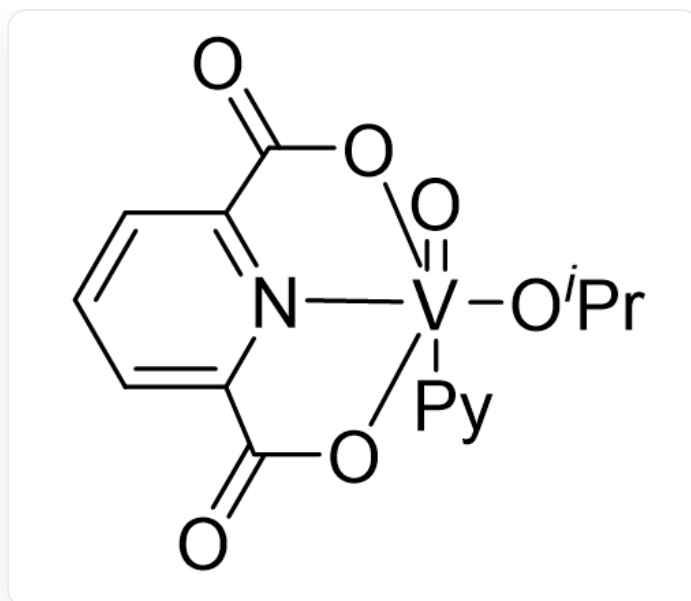


PathIII



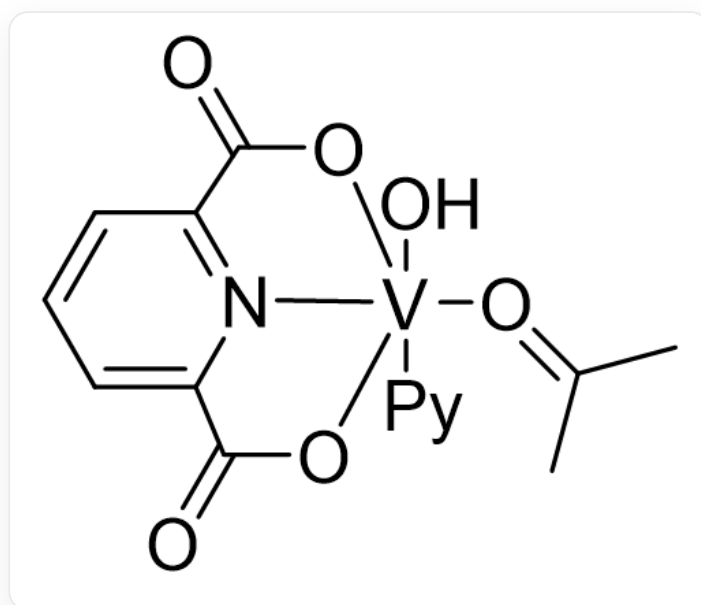
The structures of **B**, **C**, and **D** are as follows:

B:



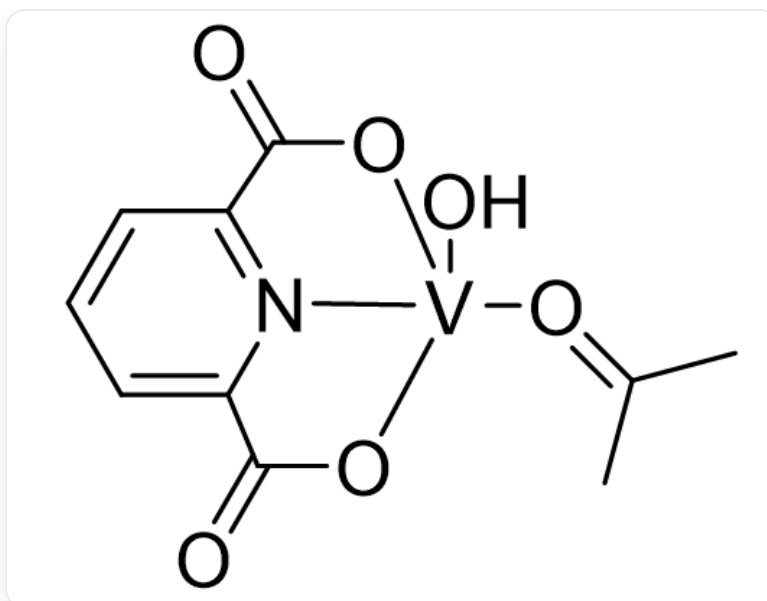
O=[V]12(OC(C)C)(C3=NC=CC=C3)[N]4=C(C(O2)=O)C=CC=C4C(O1)=O

C:



O[V]12(/O=C(C)/C)(C3=NC=CC=C3)[N]4=C(C(O2)=O)C=CC=C4C(O1)=O

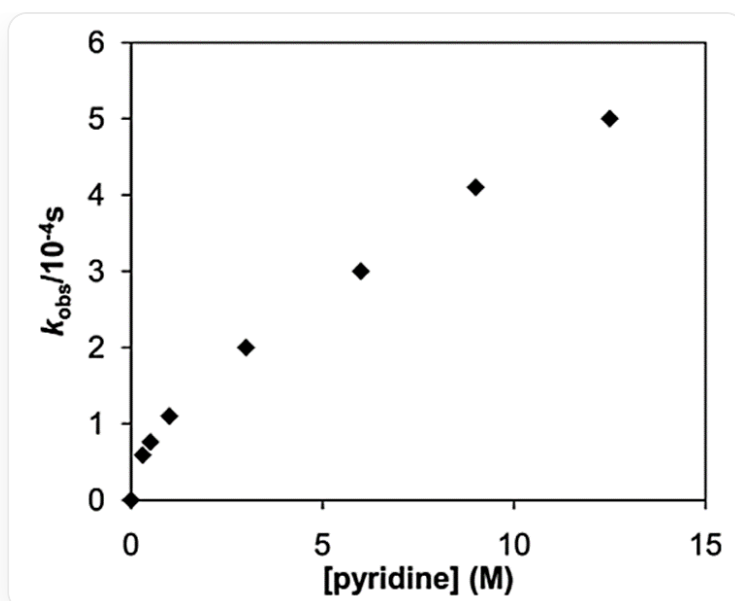
D:



O[V]12(/O=C(C)/C)[N]3=C(C(O2)=O)C=CC=C3C(O1)=O

It is known that in *PathII* and *PathIII*, one molecule of $py - d_5$ participates in the proton transfer process during the formation of **C** or **D**, while **C** and **D** are highly reactive, so their concentrations can be neglected.

Experimental measurements at $340K$ show the following variation curve of k_{obs} with $[py]$:



This image is a scatter plot of the apparent rate constant k_{obs} versus the concentration of perdeuterated pyridine $[pyridine]$. The x-axis represents $[pyridine]$ in units of M , with a range of 0 – 15, and the y-axis represents k_{obs} in units of $10^{-4}s$, with a range of 0 – 6. The estimated coordinates of the data points are

(0, 0), (0.33, 0.59), (0.49, 0.75), (0.98, 1.47), (2.95, 1.98), (5.93, 2.98), (8.93, 4.07), (12.40, 4.96).

Based on the above information, which of the following statements are **incorrect**?

- For all three reaction pathways, the reaction order with respect to $[A]$ is 1.
- PathII** has a reaction order of 2 with respect to $[py]$.
- When $[py] \gg 1/K_{eq}$, the variation of k_{obs} with $[py]$ at 340K best fits **PathI**.

A. 1

B. 2

C. 3

D. 1,2

E. 2,3

F. 1,3

G. 1,2,3

H. All three statements are correct

Answer

Correct Answer: **E**

Detailed Explanation

Let $C_{V(V)}$ denote the sum of the concentrations of all species containing only $V(V)$.

For *Path I*, the rate equation of the reaction can be expressed as:

$$r = k_1[\mathbf{B}]$$

By material balance:

$$C_{V(V)} = [\mathbf{B}] + [\mathbf{A}]$$

Additionally, the equilibrium relationship is:

$$K_{eq} = [\mathbf{B}]/[[\mathbf{A}][py]]$$

Thus, we have:

$$[\mathbf{B}] = K_{eq}[py]C_{V(V)}/(K_{eq}[py] + 1)$$

$$r = k_1K_{eq}[py]C_{V(V)}/(K_{eq}[py] + 1)$$

For *Path II*, the rate equation of the reaction can be expressed as:

$$r = k_1[py][\mathbf{B}]$$

The material balance and equilibrium relationship for *Path II* are the same as for *Path I*, so we have:

$$r = k_1K_{eq}[py]^2C_{V(V)}/(K_{eq}[py] + 1)$$

For *PathIII*, the rate equation of the reaction can be expressed as:

$$r = k_1[py][B] + k_2[py][A]$$

From the material balance of **A** and the K_{eq} relationship, we obtain:

$$[B] = K_{eq}[py]C_{V(V)}/(K_{eq}[py] + 1)$$

$$[A] = C_{V(V)}/(K_{eq}[py] + 1)$$

CHECKPOINT

0.5 PTS

$$[B] = K_{eq}[py]C_{V(V)}/(K_{eq}[py] + 1)$$

CHECKPOINT

0.5 PTS

$$[A] = C_{V(V)}/(K_{eq}[py] + 1)$$

$$r = k_1K_{eq}[py]^2C_{V(V)}/(K_{eq}[py] + 1) + k_2[py]C_{V(V)}/(K_{eq}[py] + 1)$$

Now, let's analyze each statement:

1. The three reaction paths can be expressed in terms of **[A]** as:

$$r_{PathI} = k_1K_{eq}[py][A]$$

$$r_{PathII} = k_1K_{eq}[py]^2[A]$$

$$r_{PathIII} = (k_1K_{eq}[py]^2 + k_2[py])[A]$$

Thus, for all three reaction paths, the reaction order with respect to $[A]$ is 1. This is correct.

CHECKPOINT

1 PTS

For all three reaction paths, the reaction order with respect to $[A]$ is 1. Statement 1 is correct.

2. The rate equation for *PathII* is:

$$r = k_1 K_{eq} [py]^2 C_{V(V)} / (K_{eq} [py] + 1)$$

CHECKPOINT

1 PTS

$$r = k_1 K_{eq} [py]^2 C_{V(V)} / (K_{eq} [py] + 1)$$

It is evident that $[py]$ does not follow a simple integer-order relationship. Thus, the statement is incorrect.

CHECKPOINT

1 PTS

$[py]$ does not follow a simple integer-order relationship. Statement 2 is incorrect.

3. When $[py] \gg 1/K_{eq}$, the apparent rate constants for each reaction path are:

$$PathI : k_{obs} = k_1$$

$$PathII : k_{obs} = k_1 [py]$$

$$PathIII : k_{obs} = k_1 [py] + k_2 / K_{eq}$$

CHECKPOINT

0.5 PTS

$$PathIII : k_{obs} = k_1[py] + k_2/K_{eq}$$

At high concentrations of $py - d_5$, plotting the tangent line of k_{obs} reveals that k_{obs} has a vertical intercept and exhibits a positive correlation with $[py]$ (approximately linear). Therefore, *PathIII* is the most plausible reaction pathway, and the statement is incorrect.

CHECKPOINT

1 PTS

At high concentrations of $py - d_5$, plotting the tangent line of k_{obs} reveals that k_{obs} has a vertical intercept.

CHECKPOINT

0.5 PTS

PathIII is the most plausible, so statement 3 is incorrect.

Thus, the correct answer is E (2,3).